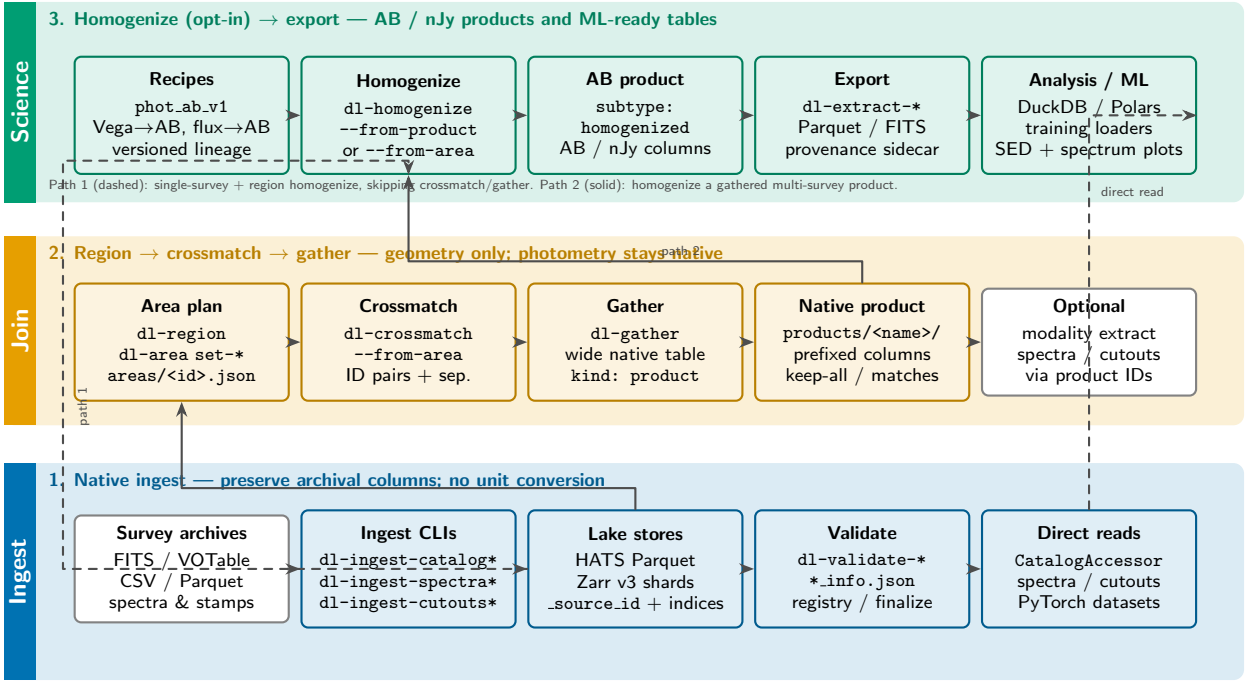
Science
3. Homogenize (opt-in) → export — AB / nJy products and ML-ready tables
Recipes
phot_ab_v1
Vega→AB, flux→AB
versioned lineage
Homogenize
dl-homogenize
--from-product
or --from-area
AB product
subtype:
homogenized
AB / nJy columns
Export
dl-extract-*
Parquet / FITS
provenance sidecar
Analysis / ML
DuckDB / Polars
training loaders
SED + spectrum plots
Path 1 (dashed): single-survey + region homogenize, skipping crossmatch/gather. Path 2 (solid): homogenize a gathered multi-survey product.
direct read
Join
2. Region → crossmatch → gather — geometry only; photometry stays native
path 2
Area plan
dl-region
dl-area set-*
areas/<id>.json
Crossmatch
dl-crossmatch
--from-area
ID pairs + sep.
Gather
dl-gather
wide native table
kind: product
Native product
products/<name>/
prefixed columns
keep-all / matches
Optional
modality extract
spectra / cutouts
via product IDs
path 1
Ingest
1. Native ingest — preserve archival columns; no unit conversion
Survey archives
FITS / VOTable
CSV / Parquet
spectra & stamps
Ingest CLIs
dl-ingest-catalog*
dl-ingest-spectra*
dl-ingest-cutouts*
Lake stores
HATS Parquet
Zarr v3 shards
_source_id + indices
Validate
dl-validate-*
*_info.json
registry / finalize
Direct reads
CatalogAccessor
spectra / cutouts
PyTorch datasets